

Liam Joseph Nolan

Senior Mechatronics Engineer | Industrial PhD Candidate



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About me

Mechatronics engineer with 6+ years in surgical and construction robotics. I take robotic systems from concept to field test: mechanical design, controls, firmware, and manufacturing. First engineering hire at Rockslide. Industrial PhD candidate (NTNU) in robot control and material sensing.

Language

English: Native
Norwegian: Basic
German: Basic

Skills

- Python
- MATLAB
- ROS 2
- Gazebo
- RViz
- micro-ROS
- Simulink
- LQR/LQI
- System ID
- C++
- Linux
- Git
- OpenCV
- GitHub Actions
- SolidWorks
- Onshape
- Inventor
- Fusion 360
- GD&T
- DFMEA
- FEA
- Ansys
- CNC Mfg
- Mastercam
- Fusion CAM
- BobCAM
- 3D Printing
- Waterjet
- Laser Cutting

ACADEMICS

Industrial PhD Candidate

Autumn 2026 – Present

NTNU, Department of Mechanical and Industrial Engineering | Field Robotics Lab
Research Council of Norway (NFR) Industrial PhD, in partnership with Rockslide
Proposed thesis: Optimizing Material Removal Rates in Robotic Stone and Concrete Cutting through Advanced Control and Material Sensing

M.Sc in Mechatronics

MCI | Innsbruck, Austria
October 2023 – July 2025

B.Sc in Aerospace Engineering

Cal Poly San Luis Obispo | USA
September 2015 – June 2019

EXPERIENCE

Senior Mechatronics Engineer Rockslide

Sept 2025–Present
Bergen, NO

National Startup Award 2024 winner: explosive-free robotic rock and concrete cutting

- First engineering hire, owning design, prototyping, and testing
- Redesigned patented water-turbine sword saw: 10× cheaper, 20→10 kg, cut depth 1→1.5 m
- Designed 3-DoF robotic sword positioner for flush, square-cornered cuts in confined spaces
- Built up lab and machine shop; installed and programmed a Haas VF-2SS CNC mill
- Sourced motion hardware and managed suppliers for machined and metal 3D-printed parts
- Debugged firmware and Modbus communication on the in-house custom control board
- Defined requirements with customers and pitched the company to investors
- Leading the company's first customer pilot project (December 2026)
- Pursuing NTNU industrial PhD on control and material sensing for the cutting platform

Master's Thesis MCI

2024–2025
Innsbruck, AT

ROS 2-based Implementation of Advanced Control for a Teleoperated Surgical Robot

- Ported a surgical teleoperation robot to ROS 2 / micro-ROS in C++
- Identified joint dynamics via step-response and closed-loop PRBS system identification
- Designed gain-scheduled LQR and LQI + feedforward controllers
- Reduced mean tracking error by 94% (pitch) and 74% (roll) vs. proportional control
- Built URDF/SDF kinematic model from SolidWorks with TF forward kinematics in RViz
- Published results as first author at IEEE ICARM 2026

Mechanical Engineer Ethicon Inc. (Formerly Auris)

2019–2023
Santa Clara, CA

Johnson & Johnson surgical robotics; early engineer on the Ottava soft-tissue surgical robot

- Project lead of patented surgeon console immobilization system
- Designed, tested, and implemented robotic subsystems for surgical robotic system.
- Diagnosed and debugged system-level mechanical and software issues in clinical settings.
- Led project to develop a mechanical immobilization solution for the surgeon console.
- Managed capital equipment packaging project as Team Lead.
- Designed and coordinated manufacturing of PCB and optical test fixtures.
- Directed repair efforts for surgical robotic systems in clinical settings.
- Supported surgical procedure development in lab environments.

Mechanical Engineering Intern Ethicon Inc. (Formerly Auris)

2018
Santa Clara, CA

- Designed and manufactured immobilization actuator for a surgical system.
- Assembled, tested, and characterized robotic arm performance.
- Tuned, characterized, and tested a custom electromechanical actuator.
- Analyzed critical tolerance stackups in custom actuator assemblies.
- Created, reviewed, and delivered part drawings for manufacturing.

Mechanical Engineering Intern Orbital ATK

2017
Goleta, CA

Aerospace and defense company (now Northrop Grumman) building spacecraft deployables

- Prototyped assembly methods for satellite deployables.
- Assembled and tested engineering development units.
- Inspected flight hardware for defects and verified assembly tolerances.
- Manufactured and assembled stiffness and thermal test fixtures.

Mechanical Engineering Intern Breedt

2015–2016
Kent, WA

Designs and builds production tooling and automation for aerospace manufacturing

- Designed, manufactured, and tested large-scale industrial tooling.
- Responsible for mechanical and electrical design of a robotic hydroponic foam cutter.
- Designed and manufactured a rubber compression mold stripper.
- Operated CNC manufacturing equipment to fabricate parts for external customers.

PUBLICATIONS & PATENTS

- **L. Nolan**, M. Messner, S. Winkler, and Y. Kim, "Implementation of an Advanced Control System for a Teleoperated Surgical Robot Using ROS 2," *IEEE International Conference on Advanced Robotics and Mechatronics (ICARM)*, 2026.
- Surgeon console immobilization system, Ethicon Inc. Patents US20230390010A1 and WO2023238053A1.

OTHER INVOLVEMENTS

FIRST Robotics:

FRC Team 2046 (President)
FRC Team 973 (Mentor)
2017 FRC World Champion

BattleBots:

Sole Engineer on Alexis
Lead Engineer on Zenith

PolySat:

Aero Team
ISX Radiation Lead

PERSONAL PROJECTS

Web Scraper:

Developed a Python script to scrape data from a local climbing gym's website to monitor its capacity. Automated the process using GitHub Actions and created a Flask app to export the data in JSON format. Dockerized the code-base and deployed it as an API endpoint on Render, while also implementing an HTML script to visualize the data on my website.

Trajectory Tracker:

Developed a Python application utilizing the OpenCV library for advanced vision processing to track, plot, and predict the trajectory of a thrown ball in video footage. The project captures real-time position, velocity, and acceleration data for each frame, providing a comprehensive analysis of projectile motion.

Delta Manipulator:

Utilized MATLAB, Simulink, and Simscape Multibody to model and simulate an IGUS Delta Manipulator. Developed end-effector trajectories using trapezoidal velocity profiles, with inverse kinematics to compute joint positions and forward kinematics to update the Simscape model.

Weather Frog:

Built a Python script that displays the current weather of a chosen city in a Linux terminal, fetching data from the OpenWeatherMap API and presenting it in a formatted table.

TurtleBot Factory:

Designed custom hardware for the TurtleBot4 to facilitate the reception and delivery of syringe blocks. Developed custom ROS2 packages for multi-robot navigation and docking, utilizing LIDAR and the Navigator for collision avoidance and efficient robot pathing.

Turtlesim Edge Detection:

Developed a custom C++ node for ROS 2 that spawns a simulated TurtleBot at a random orientation. The robot autonomously drives forward, utilizing edge detection to identify boundaries, then executes a 90-degree turn and continues moving.

Dice Sorting Robot:

Developed an automated sorting system using an ABB robot arm and Cognex camera to track and sort dice. Move instructions were programmed in RAPID, with vision processing handled by a custom Cognex application.

Climbing Data Tracker:

Created a tool to log and analyze climbing training data recorded in markdown via Obsidian. A Python parser feeds a Dockerized backend on Render that powers a web dashboard with interactive graphs.